

Aoun: Security Verification Report

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Two-way LoRa distress and tracking system · Team: D. Alanazi (Leader), R. Aldosari, J. Alsharif, H. Alhejjei, A. Alqarni · Digital Technical College for Girls, Riyadh
v2 adds on-device results from the assembled LILYGO T-Beam V1.2 boards (7 October 2026). It replaces v1 (September 2026, PC tests only).

1. Scope

This report verifies the message-security layer of Aoun. The layer is one block of C++ code, identical in the victim and rescue firmware. It was tested in two ways: (a) compiled for a PC host against Mbed TLS 2.28.8, the same 2.28 line built into the ESP32 Arduino core, and run through 10 automated test cases; (b) run on two real T-Beam boards, including deliberate attack tests over the air.

2. Design

- Cipher:** AES-128-GCM authenticated encryption, 128-bit key shared by paired devices, 16-byte authentication tag. Key generated on the ESP32 hardware random number generator; never published.
- Nonce:** sender ID + 32-bit sequence number (boot counter kept in flash × message counter), so a nonce is never reused.
- Header** (version, sender ID, sequence) is authenticated as associated data. Packet: [version 1][sender ID 2][sequence 4][ciphertext N][tag 16].
- Replay protection:** the receiver accepts only sequence numbers higher than the last one seen from that device (checked after the tag, so forgeries cannot block real messages).
- Whitelist:** only registered device IDs are accepted.

3. PC test results: 10 / 10 passed

Test case	Result
Valid SOS decrypts correctly	PASS
Location not visible in the radio packet	PASS
Same packet sent again is rejected (replay)	PASS
One flipped bit in ciphertext is rejected	PASS
Modified header (sequence) is rejected	PASS
Packet made with the wrong key is rejected	PASS
Unregistered device ID is rejected	PASS
ACK + message reaches victim, matches last SOS	PASS
Packets after victim reboot still accepted	PASS
Truncated packet is rejected	PASS

A packet from the suite was also decrypted with an independent AES-GCM library (Python *cryptography*), confirming standard GCM output.

4. On-device results (2 × T-Beam V1.2)

Test over the air	Result
Encrypted SOS received and verified	PASS
Encrypted ACK verified by victim unit	PASS
Automatic retry (3 × every 10 s) until ACK	PASS
Wrong-key SOS rejected	5 / 5
Unregistered-device SOS rejected	3 / 3
Location readable in captured packet	No
Genuine SOS accepted again after attacks	PASS

Indoor link test before encryption (unencrypted PING every 3 s, SF9 / BW 125 kHz / 10 dBm): 100% delivery at 2 m, next room, through a concrete floor and outdoors in the yard; RSSI −31 to −80 dBm, SNR 9.5 to 11.25 dB.

5. Packet size and airtime

37 bytes SOS packet (encrypted), confirmed on hardware	62 bytes ACK + 34-character message, confirmed on hardware	≈ 0.34 s SOS airtime at SF9 (2.5 s at SF12)	23 bytes security overhead per packet
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Airtime: Semtech LoRa time-on-air formula (BW 125 kHz, CR 4/7, 8-symbol preamble, explicit header, CRC on).

6. Limitations and next steps

- The rescue unit keeps the last accepted sequence number in RAM; after a rescue-unit restart one old packet could be accepted once. Planned fix: store it in flash.
- Next: field tests in open terrain (range at 0.5 / 1 / 2 / 5 km, delivery rate over 50 SOS per distance, SOS-to-confirmation time, battery life with solar), after confirming the permitted band and power with CST.